

Alternate FLT proof + Challenge Problem solution

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1 Fermat's Little Theorem

Theorem 1. *If p is prime and $\gcd(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod{p}$.*

Proof using Group Theory. Without loss of generality assume $1 < a < p$. In particular, a is a member of the finite set $\{1, \dots, p-1\}$.

Let A be the set of first p powers of $a \pmod{p}$, i.e. $A = \{a^1, a^2, \dots, a^p\}$, all $\pmod{p}$. Note that $|A| = p$, but $|\{1, \dots, p-1\}| = p-1$. By the Pigeonhole Principle, there must be distinct positive integers $i, j \leq p$ s.t. $a^i = a^j$. Since a^i has an inverse, we can multiply both sides by $a^{-i} = (a^i)^{-1}$ to get $1 = a^{j-i}$. Set $k = j - i$, and without loss of generality assume that k is the least positive integer so that $a^k = 1$.

Claim: k is a factor of $p-1$. In group theory this claim is a special case of a more general theorem called Lagrange's Theorem.

Proof of Claim: If $k = p-1$ then we're finished, so we will just consider the case where $k < p-1$. Let b be an arbitrary element of $\{1, \dots, p-1\}$ not in A (there exists such an element b because $|A| = k$). Define $Ab = \{ba^n \mid n \leq k\}$ (in group theory sets like this are called *cosets*). Note that the function $f(x) = bx \pmod{p}$ is a bijection between $\{1, \dots, p-1\}$ and itself, and restricting f to A gives us a bijection between A and Ab . Hence, A and Ab have the same cardinality. But also, they have no elements in common; since A is closed under multiplication and $b \notin A$ it follows that $a^n b \notin A$ either (otherwise we could get $b \in A$ by taking inverses). Thus, $\{1, \dots, p-1\}$ can be partitioned into finitely many cosets, all with the same cardinality k . Thus, $p-1 = |\{1, \dots, p-1\}|$ is a multiple of k .

To finish, using our claim we can write $p-1 = kt$ for some integer t , and so $a^{p-1} = a^{kt} = (a^k)^t = 1^t = 1$. $\square$

2 Challenge Problem

We will define $\mathbb{N}$ as the set of nonnegative integers, and $\mathbb{N}^{<\omega}$ as the set of all finite sequences of nonnegative integers. In other words, $\mathbb{N}^{<\omega} = \bigcup_n \mathbb{N}^n$.

Theorem 2. *The set of all finite sequences of nonnegative integers is countable.*

Proof 1, using the fact that countable unions of countable sets are countable. Let $\mathbb{N}$ be the set of all nonnegative integers. We know that there is a bijection between $\mathbb{N}^2$ and $\mathbb{N}$. Using this, we can inductively show that for any $n \geq 2$, $\mathbb{N}^n$ is countable. For $n = 2$ we just did it. For $n > 2$, we can factor $\mathbb{N}^n$ into $\mathbb{N}^{n-1} \times \mathbb{N}$, and using a bijection between $\mathbb{N}^{n-1}$ and $\mathbb{N}$ we can make a new bijection between $\mathbb{N}^n$ and $\mathbb{N} \times \mathbb{N}$, which we already know is countable.

Then, since $\mathbb{N}^{<\omega} = \bigcup_n \mathbb{N}^n$ is a countable union of countable sets, it is countable. $\square$

Of course, to complete the proof we have to show that the countable union of countable sets is countable. To do this, we have to use an axiom called the *axiom of choice*. This one was a little controversial in set theory when it was first formulated, but it turns out that it's necessary to prove things like this. I'll mention explicitly when we use it in the proof.

Lemma 1. *The countable union of countable sets is countable.*

Proof. Let $A = \bigcup_{i \in \mathbb{N}} A_i$ be a countable union of countable sets. Without loss of generality we will assume that if $i \neq j$, then $A_i \cap A_j = \emptyset$ (if this isn't the case, we can redefine A_i as $A_i \setminus \bigcup_{j < i} A_j$). We will make a bijection between A and $\mathbb{N}^2$. For each n , we **choose a bijection** f_n between A_n and $\mathbb{N}$ (this is the step that uses the Axiom of Choice). Then, we define $g(m, n)$ as $f_m(n)$.

First, notice that g is a well-defined function from $\mathbb{N}^2$ to A . To see that g is one-to-one, let $(m, x), (n, y) \in \mathbb{N}^2$ be arbitrary, and assume $g(m, x) = g(n, y)$. By definition, we have $f_m(x) = f_n(y)$. Since we assumed the A_i s are disjoint, we must have $m = n$; otherwise they would map to different elements. So we get $f_n(x) = f_n(y)$. But since f_n is a bijection, we get $x = y$.

To check that g is onto, let $x \in A$ be arbitrary. Then $x \in A_i$ for some i . Since f_i is a bijection between A_i and $\mathbb{N}$, we get that there is an n with $f_i(n) = x$. By definition, this means that $g(i, n) = x$. Hence, g is onto. $\square$

If you happen to feel qualmy about the Axiom of Choice, there is also an alternative proof of Theorem 2 that does not use it.

Proof 2, more direct by building an explicit bijection. We build an explicit bijection between $\mathbb{Z}^+$ and $\mathbb{N}^{<\omega}$. Let p_i denote the i th prime number. For $n \in \mathbb{Z}^+$, we find the prime factorization of n as $2^{a_1} \cdot 3^{a_2} \cdots p_n^{a_n}$. Then we define $f(n)$ as $(a_1, \dots, a_{n-1}, a_n - 1)$ (we had to use $a_n - 1$ at the end because a_n is always positive, but the prior a_i s might be 0 for $i < n$).

To check that f is a bijection, first note that f is well-defined by the Fundamental Theorem of Arithmetic. To check that f is one-to-one, suppose $f(x) = f(y) = (a_1, \dots, a_n)$. By definition of f , we have $x = y = 2^{a_1} \cdot 3^{a_2} \cdots p_n^{a_n+1}$. Now, to check that f is onto. Let $(a_1, \dots, a_n)$ be a finite sequence of nonnegative integers. Then, the number $x = 2^{a_1} \cdot 3^{a_2} \cdots p_n^{a_n+1}$ exists, and $f(x) = (a_1, \dots, a_n)$. $\square$

While prime factorization is the most straightforward bijection between $\mathbb{Z}^+$ and $\mathbb{N}^{<\omega}$, it is also possible to make one using the Chinese Remainder Theorem. Kurt Gödel used this one when he proved his famous *Incompleteness Theorems*,

which showed that there is a fundamental limit to what we can prove in any formal axiomatic system. The Incompleteness Theorems have inspired a lot of philosophical speculation about the nature of mathematics and the human mind.

I will leave the bijection via the Chinese Remainder Theorem as an extra challenging exercise, in case someone wants to try it.